TP2 (JavaScript Version)

INF8808 : Data Visualisation

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Objectives

The goal of this lab is to create an interactive grouped bar chart using open data in CSV format.

Before beginning, we recommend you have completed the following readings and practice exercises :

Readings :	Chapters 6, 7 and 8 of Scott Murray's book
Exercices :	• Chapter 3 – 1, 2, 3 • Chapter 4 – 1, 2 • Chapter 5 – 1, 2

Introduction

A bar chart is a type of data visualisation which represents data using rectangles whose lengths are proportional to the values they represent. In a grouped bar chart, the bars are divided into groups using their position on the chart. Across each group, each bar's color denotes its subgroup. Grouped bar charts are commonly used to compare the values of each subgroup across different categories.

In this lab, you will implement a grouped bar chart using data from Shakespeare's play, *Romeo and Juliet*. The data was extracted from Kaggle, a popular data science website [1]. The dataset was then modified into a version to be used in this lab. It contains data on the entire text of *Romeo and Juliet*.

Description

In this lab, you will have to complete the JavaScript code using D3 in order to display a grouped bar chart displaying data on how many lines each player has in each act of the play. To make the chart interactive, you will also implement the code to display an informational tooltip when each bar is hovered. Additionally, there will be a legend indicating the color corresponding to each player displayed on the chart.

Each group will contain as subgroups one bar for each of the top 5 players who utter the most lines in the play. The other players will be grouped together into their own category "Others", containing their summed line count.

The following subsections present the different parts that you will have to complete for this lab. While you code, we recommend completing the data processing first, followed by the implementation of the bar chart itself. The next two parts, the legend and the tooltip are independent of each other.

File Structure

To complete this work, you will need to fill the various `TODO` sections in the files from the archive provided for the lab. The comments in the code explain in more detail the steps to take.

In this lab, we provide you with an archive containing 6 JavaScript files used to accomplish the desired visualization :

- `index.js` : This file represents the entry point to the code and orchestrates the various steps needed to realise the visualization. It does not need to be modified.
- `scripts/helper.js` : This file contains some basic functions needed to display the visualization. It does not need to be modified.
- `scripts/legend.js`
- `scripts/preprocess.js`
- `scripts/tooltip.js`
- `scripts/viz.js`

Dataset

The dataset is located in the `src/assets/data/` directory in the archive provided for the lab. The dataset contains the following columns :

- `Act` : This column represents the act in which the line was uttered.
- `Scene` : This column represents the scene in which the line was uttered.
- `Line` : This column tracks the number of the line for the given act and scene.
- `Player` : This column contains the name of the player who uttered the line.
- `PlayerLine` : This column contains the content uttered by the given player during the given act and scene at the line with the given number.

Note : As a reminder, the dataset represents data from a play, which are usually divided into large chunks called “acts”. In this play, there are 5 acts. In turn, each act is divided into scenes, which may contain any number of lines, presented in the order in which they are uttered during the play. Also note that by “player” we are referring to a character played by an actor.

Data preprocessing

To begin, you will have to preprocess the data we provide you. The data contained in the CSV file is raw, so it is necessary to reorganize certain parts of it so they can be properly used by the D3 library. To do so, you need to complete the file `scripts/preprocessing.js`.

More precisely, you will have to complete these steps :

1. Update the names of the players in the dataset so they follow correct capitalization (function `cleanName`)

2. Implement a function to find the names of the 5 players with the most lines in the play (function `getTopPlayers`). Only these players will be directly displayed in the grouped bar chart to avoid overcrowding.

3. Group and restructure the data into a nested structure to be used by D3 (function `summarizeLines`)

4. Modify the structure to include an "Other" category, which contains the sum of lines in that act of players not belonging to the top 5 for the play (function `replaceOthers`)

To help validate your work, Figure 1 illustrates a portion of the resulting data structure.

```
[
  {
    "Act": "1",
    "Players": [
      {
        "Player": "Benvolio",
        "Count": 34
      },
      {
        "Player": "Romeo",
        "Count": 50
      },
      {
        "Player": "Nurse",
        "Count": 20
      },
      {
        "Player": "Juliet",
        "Count": 16
      },
      {
        "Player": "Mercutio",
        "Count": 11
      },
      {
        "Player": "Other",
        "Count": 105
      }
    ]
  },
```

Figure 1 : Extract of preprocessed data

Bar chart

For this second part, you will have to implement the main part of the data visualization. First, you will have to update the scales used to position the elements in the visualisation. Second, you will use these scales to append and position the groups and subgroups to the grouped bar chart. To do so, you need to complete the file `scripts/viz.js`.

These are the steps you will have to accomplish for this part :

1. Set the domain and range of the scale determining the x position of each group of bars (function `updateGroupScale`)

2. Set the domain and range of the scale determining the length of each bar (function `updateYScale`)

3. Generate SVG groups which will contain each group of bars (function `createGroups`)

4. Append bars in each group with the appropriate height, position and color (function `drawBars`)

Figure 2 illustrates what the bar chart should look like after this step.

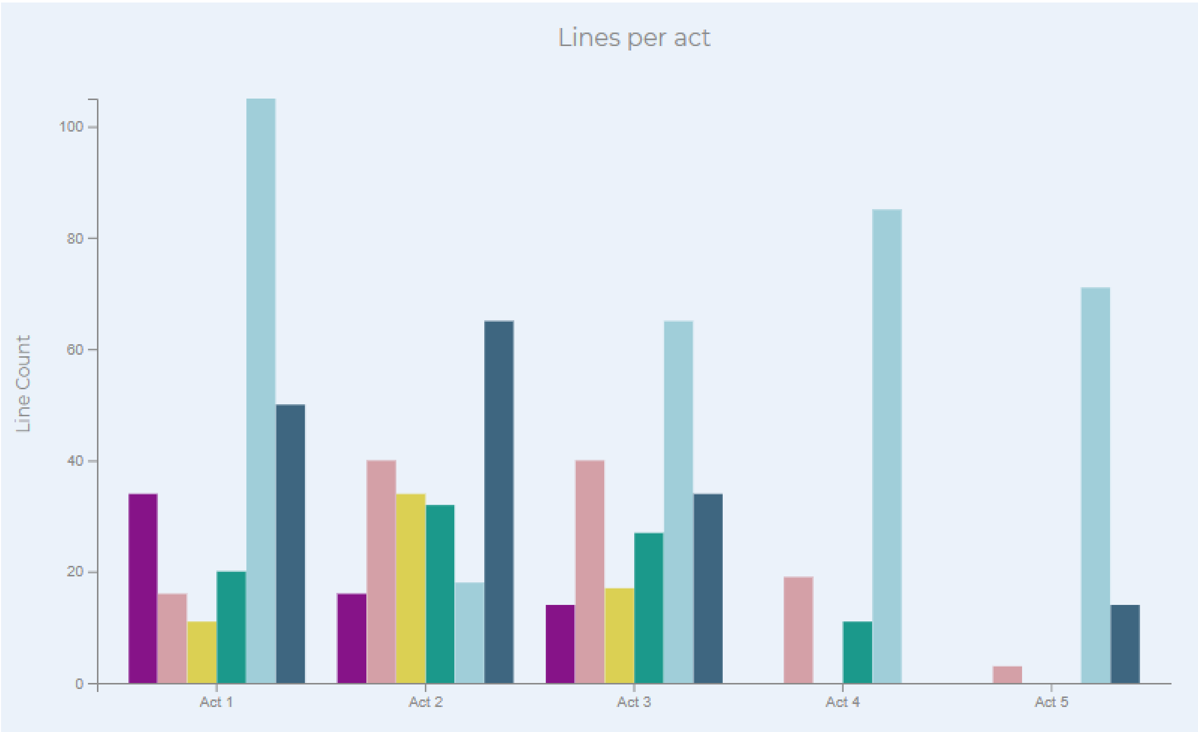

Figure 2 : The grouped bar chart

Legend

For this third part, you will generate a legend in the webpage's footer. Take a moment to notice in the provided HTML and CSS files some elements that may be helpful with positioning and appearance of the legend. To complete this part, you need to complete the file `scripts/legend.js`. The entirety of the code for this section can be written in the function `draw`. Make sure the colors in the legend match the ones in the bar chart for each player.

The resulting legend should look like the one in Figure 3.

Figure 3 : The legend in the page footer

Tooltip

For this fourth part, you will generate a tooltip which appears over each bar when it is hovered which summarizes information associated with that bar. The tooltip should contain the player name and line count for a given act. Make sure to style the title as specified in the comments in the code. To complete this part, you need to complete the file `scripts/tooltip.js`. The entirety of the code for this section can be written in the function `getContents`. Make sure you also handled the tooltip when you created the bars for the chart.

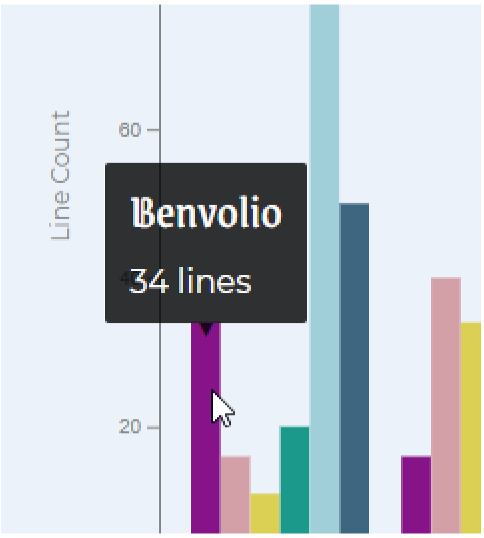

Figure 4 : The grouped bar chart with a tooltip on hover

In Figure 4, you can see the appearance of the tooltip when a bar is hovered by the cursor.

Submission

The instructions for the submission are :

1. You must place your project code in a compressed ZIP file named [**Matricule1_Matricule2_Matricule3.zip**]

2. The lab must be submitted before [**May 21st 23:59**]

Evaluation

Overall, your work will be evaluated according to the following grid. Each section will be evaluated on correctness and quality of the work.

Requirement	Points
Data preprocessing	6
Bar Chart	8
Legend	3
Tooltip	2

Requirement	Points
Overall quality and clarity of the submission	1
Total	**20**

References

[1] L. Larsen, "Shakespeare plays," Kaggle. Available: https://www.kaggle.com/kingburrito666/shakespeare-plays. [Accessed 30 07 2020].